ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory

Siru Ouyang, Jun Yan, I-Hung Hsu, Yanfei Chen, Ke Jiang, Zifeng Wang, Rujun Han, Long T. Le, Samira Daruki, Xiangru Tang, Vishy Tirumalashetty, George Lee, Mahsan Rofouei, Hangfei Lin, Jiawei Han, Chen-Yu Lee, Tomas Pfister

Paper version: arXiv 2509.25140v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Agents discard what their interaction history could teach and repeat past errors.*

Stored memory is raw trajectories or **success-only** routines.

Motivation: three gaps in agent memory

1. **Raw trajectories** are long and noisy.
2. **Success-only routines** leave failures unused.
3. **No higher-level patterns** in memory.

Evaluation setup

WebArena · 684 tasks | Mind2Web · 1341 (offline) | SWE-Bench-Verified · 500 | Gemini-2.5 flash / pro, Claude-3.7 | 30-step cap, k = 1 | MaTTS k = 5

- **WebArena is online;** Mind2Web is offline.
- **Streaming:** tasks arrive one by one, **no ground truth**.

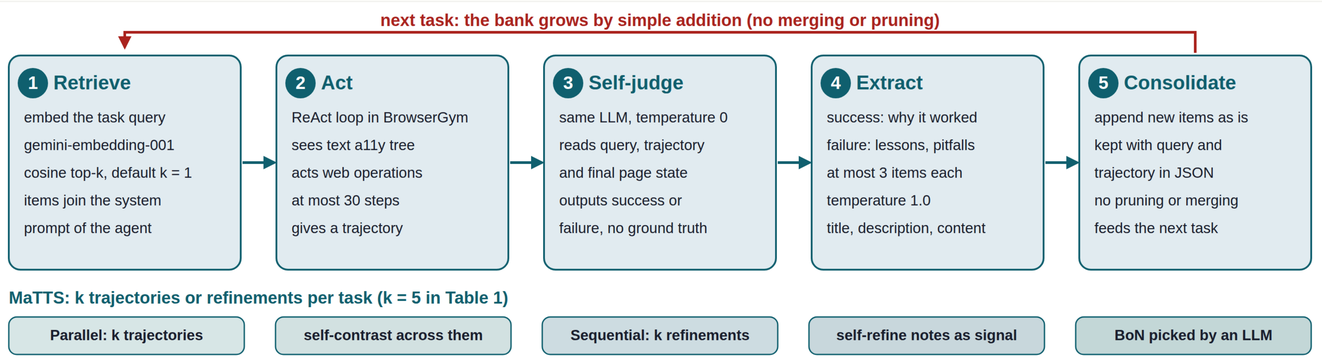
Our read (CUA Radar)

66	★★	online
score of 100	band	evidence level

- **Reports:** online WebArena eval; Synapse and AWM baselines; steps and tokens; failure ablation; judge accuracy; code.
- **Flags:** AP4, self-judge not validated two-sided. Not reported: seeds, variance, held-out tasks.
- Pattern C19. Authors' numbers, not rerun.

Method: retrieve, act, self-judge, distil, append; MaTTS scales it

- **Sees / acts:** text accessibility tree in; web operations (bash for SWE) out.
- **Memory item:** title, one-line description, content; at most 3 per trajectory.
- **Signal:** the agent's own LLM judge labels success or failure; no ground truth.



$$a_{t+1} = \pi_{\mathcal{L}}(o_{0:t}, a_{0:t}; \mathcal{M}, \mathcal{A})$$

Policy conditioned on the memory M; retrieved items join the system prompt

Key results: WebArena overall success rate (%)

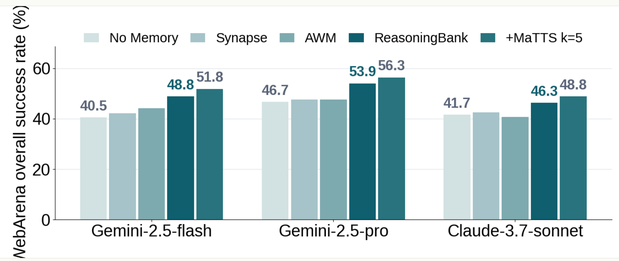


Table 1, redrawn: 684 tasks, online, pass@1; no std or CI reported.

+8.3 / +7.2 / +4.6 over No Memory (flash / pro / Claude)

SO WHAT → *Rows are pass@1; +MaTTS uses 5 parallel trajectories per task, the others one.*

Ablation: memory source, WebArena-Shopping SR (%)



Fig. 7 values quoted in the text; Gemini-2.5-flash, Shopping subset (187 tasks).

SO WHAT → *Adding failures lifts ReasoningBank (46.5 to 49.7) but not AWM (44.4 to 42.2); Synapse gains little (40.6 to 41.7). The self-judge is 72.7% accurate against ground truth on this subset; with simulated judge accuracy of 70% to 90%, the paper reports similar success rates.*

Headline Numbers

48.8%
WebArena SR, 2.5-flash
vs 40.5% No Memory, same backbone

53.9%	Gemini-2.5-pro
46.3%	Claude-3.7
51.8%	flash + MaTTS k=5

The paper's own controls

CONTROL (FLASH UNLESS NOTED)	REASONINGBANK	COMPARED WITH
Offline Mind2Web SR t/s/d	4.8 / 2.3 / 1.6	No Mem 3.3 / 1.7 / 1.0
SWE-Bench-V resolve fl/pro	38.8 / 57.4	No Mem 34.2 / 54.0
Gemma-3-12B, Shopping SR	24.1	No Mem 17.1, AWM 21.4
Tokens per task, total	53054.5	No Mem 50847.4

Takeaway

Strategy-level memory from self-judged successes and failures lifts WebArena success on three backbones; MaTTS adds more at k = 5.

judge accuracy 72.7% | Gemma-3-12B: 17.1→24.1 | MaTTS k=1→5, Shopping: 49.7→55.1

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Paper | Code